

INTL Terminal

Executive System Review & Cron Health Audit

Prepared	14 June 2026
Repository	github.com/git0ST/automation (branch: main)
Scope	India-first quantitative portfolio system — data pipeline, calibrated decision engine, auto-mode paper trading, self-learning loop
Audience	Owner / Engineering / Investment review

OVERALL STATUS	ARCHITECTURE & LEARNING: SOUND DATA PIPELINE: ACTION REQUIRED
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This report opens with the requested cron / data-gap audit (a critical finding), then documents the system, its India-first investment layer, the self-learning loop, and prioritized next steps. All figures are drawn from verified session data; where the live environment could not be re-queried (cron environment offline at time of writing), values are labelled *last-verified*.

1. Cron Health Audit — Critical Finding

■ **CRITICAL:** Scheduled data collection is failing silently

Root cause. The Python environment is reached via ~/anaconda3, which is a **symlink to a removable SanDisk SSD** (/Volumes/SSD1TB). When the SSD is unplugged or unmounted at the scheduled cron windows (06:00 / 12:00 / 17:00 IST), the interpreter path does not exist and the run aborts instantly under set -e — with no alert.

Evidence (logs). Every scheduled run on 12–13 June produced only:

```
run_digest.sh: line 23: /Users/shivamthakur/anaconda3/envs/automation/bin/python: No such file or directory
```

Window	Completed-stage runs	python-not-found failures	Status
27–31 May	up to 4/day	0	healthy
31 May → 11 Jun	0 (machine reboot dropped job)	—	11-day gap
11 Jun	4	2	partial
12 Jun	0	3	total failure
13 Jun	0	3	total failure

Why it matters

Calibration, weight-learning and the paper-trading book all depend on uninterrupted collection. Each silent gap starves the very mechanisms that make the system accurate. The earlier 11-day outage already cost more measured accuracy than any model defect to date.

Mitigation shipped this session

- **Mount-aware, fail-loud runner.** The runner now searches a candidate interpreter list (internal disk first, SSD last), attempts `diskutil mount SSD1TB` if missing, and on failure writes a `CRON_ALERT`. Log line and exits 70 — a gap can never again pass unnoticed. [commit cfec854]
- **Reboot resilience confirmed.** Both `launchd` jobs carry `RunAtLoad=true`, so a restart resumes collection.

RECOMMENDED PERMANENT FIX

Relocate the Python environment off removable media onto the internal disk (**43 GB free** — ample). This makes the schedule immune to the SSD being unplugged. Alternative: keep the SanDisk SSD permanently connected and disable its auto-unmount.

Reference file: [run_digest.sh](#)

2. System Architecture

Three independent runtimes share one calibrated decision engine — no logic is duplicated.

Runtime	Role	Writes
Headless cron (Mac · launchd)	16-source pipeline → US + India scans → predictions → outcome correlation → weight tuning → paper trading	Supabase (service key)
Streamlit Cloud app	Read-only UI behind a password gate; India Invest, India Intraday, My Plan, Paper Trading, Track Record	Reads (anon key)
Offline ML	Local feature corpus (44 features/row) + daily labeling job; first supervised model	Local JSONL

Decision engine — 6-step calibrated pipeline

Each step is grounded in established quant practice (file: [streamlit/_stock_analysis.py](#)):

1	Regime-conditional weights	weights keyed to the live market regime
2	5-signal weighted vote	technical · quant factors · analyst · sentiment · sector
3	Agreement × strength	corroboration raises, disagreement cuts conviction
4	Volatility targeting	conviction scaled inversely to realized vol
5	Systemic-risk haircut	US SRS / India VIX trims conviction in stress
6	Empirical calibration	stated confidence pulled to observed hit-rate

Beyond the directional call: a holding-horizon inference, an explicit AVOID screen, and quarter-Kelly position sizing under a 60% gross-exposure cap with volatility-scaled stops.

3. India-First Investment Layer

Portfolio growth is a positional problem, so the full engine now runs on the NSE with India-native inputs: yfinance NSE fundamentals (P/E, ROE, growth, margins, 30–40-analyst consensus per name), India VIX mapped onto the risk haircut, and the NIFTY SMA50/200 trend regime gating shorts.

Surface	Purpose	Source
My Plan	Beginner cockpit: core-satellite allocation with live evidence gates + system-health grade	0_My_Plan.py
India Invest	NIFTY 50 positional BUY list, factor grades, analyst upside, rupee Kelly sizing, holdings review	0_India_Invest.py
India Intraday	Live NSE setups: VWAP, opening-range breakout, RVOL, ATR stops, session discipline	0_India_Intraday.py
Paper Trading	Auto-mode virtual book: equity curve, open risk, closed ledger, adaptation state	B_Paper_Trading.py

Live India read (last-verified, 13 Jun)

Regime: **NIFTY BEAR · India VIX 15.6** — the engine correctly compressed conviction (no name above 55%), ranking quality-at-a-discount as best risk/reward:

Name	Quant	Analyst upside	Analysts	Note
ICICI Bank	B (65)	+31%	40	strongest consensus gap
Shriram Finance	B (70)	+30%	32	best quant score
NTPC	B (63)	+23%	28	defensive utility
Adani Ports	B (65)	+7%	25	only strong momentum (+28% 3m)
Axis Bank	C+ (58)	+23%	39	financials theme

Sizing discipline: stage into the top names at the page's quarter-Kelly sizes, honor volatility-scaled stops, keep total deployment under the 60% gross cap, and do not short in a bear trend (the system measured its own bearish calls at a 3% hit rate). Core index SIP remains the primary stable-growth engine, untouched by signal quality.

4. Self-Learning & Market Adaptation

The loop was executed this session: **240 predictions settled** against realized 7-day returns, weights re-tuned and activated (v1.1-learned), and empirical calibration verified live. File: [shared/learning_loop.py](#).

Calibration — stated confidence vs measured reality (last-verified)

Band / direction	Settled	Hit rate	Engine response
Bullish 60–69%	45	80%	sweet spot — trusted
Bullish 70–79%	20	75%	well-calibrated, untouched
Bullish 80–100%	15	0%	over-confident band → cut to ~62%
Bearish (all)	67	3%	anti-predictive → conviction slashed

Net effect: the calibration is self-correcting — the long book delivered **+3.3% vs SPY -2.0%** ($\approx +5.3\text{pp}$ alpha) over the window, while the system now declines the call types it measured itself failing at. The very next scan logged 9 selective calls instead of ~50.

Auto-mode paper trading + adaptation

Every cron pass the system now **acts on its own calls virtually** — opening positions, managing stop / target / time-stops, and recording realized P&L (file: [shared/paper_trader.py](#)). First live cycle opened 4 positions (JNJ, BAC, LIN, MRK) from the calibrated calls.

- **Regime-flip de-risking** — NIFTY trend tracked across runs; a flip triggers a 3-day half-Kelly window with a raised entry bar.
- **Volatility brake** — India VIX > 25 forces a 2-day de-risk.
- **Self-throttling entry bar** — rolling 20-trade hit-rate moves the confidence threshold (bounded 50–70%): trade less when cold, press when hot.

Supervised ML: a first model trains on the 1,188-row labeled corpus (file: [scripts/train_model.py](#)). It is **deliberately gated out of production** — with only a few distinct collection days the purged test is empty and the high AUC reflects memorization, not skill. A ≥ 30 -distinct-day gate blocks live wiring. This is exactly why fixing the cron gap (Section 1) is the top priority.

5. Artifacts & Source Links

All paths are clickable and resolve to the main branch.

File	What it is
shared/india_swing.py	NIFTY 50 positional engine (fundamentals + analyst consensus + India VIX)
shared/india_market.py	NSE universe + IST trading-session state machine
shared/paper_trader.py	Auto-mode paper book + market-shift adaptation controller
shared/system_health.py	Uptime / learning / paper health → GREEN-AMBER-RED grade
shared/learning_loop.py	Outcome correlation, weight tuning, calibration map
shared/data_lake.py	44-feature ML corpus + labeling
shared/breeze_client.py	ICICI Breeze live quotes + hard-gated order tickets
streamlit/_stock_analysis.py	6-step calibrated prediction engine
scripts/train_model.py	First supervised model (purged eval, live-wiring gate)
scripts/label_features.py	Daily labeling job (forward-return labels)
supabase/migrations/015_paper_trading.sql	Paper-trades table (RLS: anon-read / service-write)
run_digest.sh	Cron runner — now mount-aware & fail-loud

Key commits this session (most recent first)

Hash	Change
cfec854	cron runner mount-aware + fails loudly (this audit's fix)
7604caa	My Plan page — core-satellite plan with evidence gates + health
4f56ab2	auto-mode paper trading with market-shift adaptation
58611b1	India-first portfolio layer — swing engine, daily cron, Invest page
349eefa	execute the learning loop — settle outcomes, learn weights, first model
ad3da2c	ICICI Breeze integration — live quotes + hard-gated order tickets

6. Prioritized Next Steps

P0	Eliminate the data gap (owner action). Relocate the conda env to internal disk (43 GB free) OR keep the SanDisk SSD permanently connected. Until then, the mount-aware runner will alert via CRON_ALERT.log instead of failing silently.
P0	Keep the Mac awake during cron windows (06:00 / 12:00 / 17:00 IST). Uptime is the single largest driver of accuracy — the My Plan health strip now grades this GREEN/AMBER/RED.
P1	Run Supabase migration 015 (paper_trades). Makes the auto-mode paper book visible in the cloud app; the cron already writes it to a local fallback.
P1	Static IP → ICICI Breeze keys. Unlocks real-time NSE pricing and a HOLD/ADD/TRIM review of your actual holdings. Integration is built and hard-gated.
P2	Let the system prove itself (~3–4 weeks), then decide. Start the core index SIP now (needs no proof). The satellite stays on paper until the My Plan evidence gates pass; do not change the engine mid-measurement so the paper book's verdict stays attributable.

Bottom line: the decision engine, learning loop and India-first investment layer are sound and measurably self-improving. The one thing standing between this system and stable, accurate output is uninterrupted data collection — now made visible and recoverable, and one owner action away from permanent.